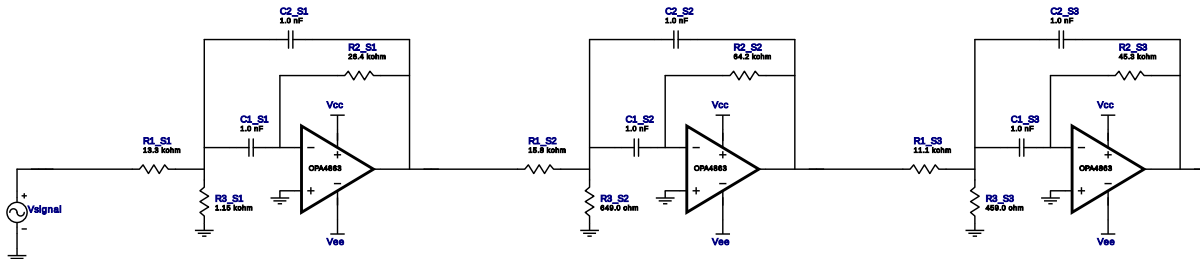


## Filter Design Report

Design : Bandpass Filter - 6th order Butterworth  
Design ID: 10



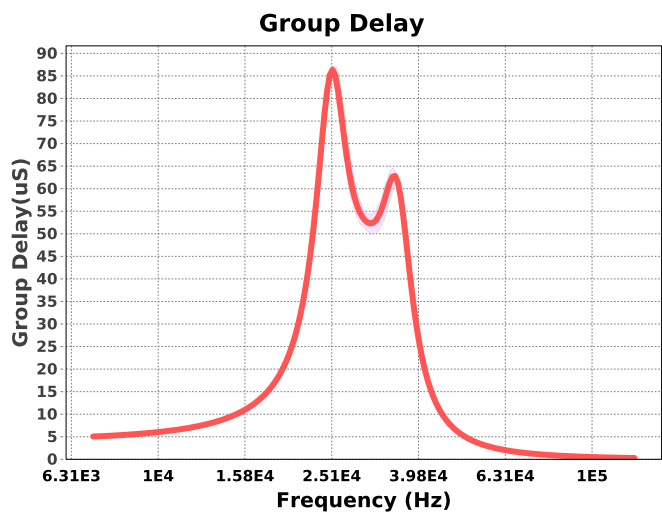
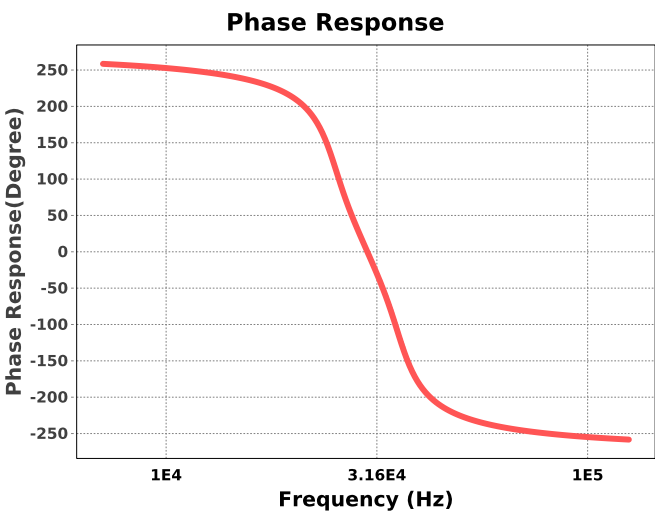
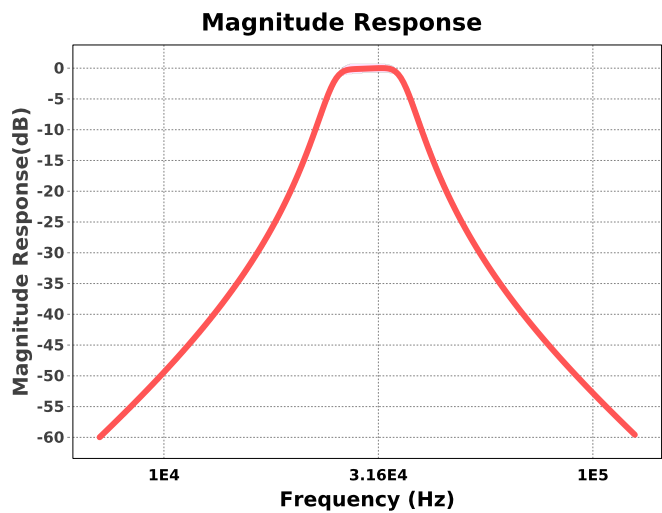
## Electrical BOM

| #   | Name  | Manufacturer           | Part Number | Properties                                     | Qty |
|-----|-------|------------------------|-------------|------------------------------------------------|-----|
| 1.  | A1_S1 | Texas Instruments Inc. | OPA4863     | GbwTyp= 50MHz<br>VccMax= 12.6V<br>VccMin= 2.7V | 1   |
| 2.  | A1_S2 | Texas Instruments Inc. | OPA4863     | GbwTyp= 50MHz<br>VccMax= 12.6V<br>VccMin= 2.7V | 1   |
| 3.  | A1_S3 | Texas Instruments Inc. | OPA4863     | GbwTyp= 50MHz<br>VccMax= 12.6V<br>VccMin= 2.7V | 1   |
| 4.  | C1_S1 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 5.  | C1_S2 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 6.  | C1_S3 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 7.  | C2_S1 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 8.  | C2_S2 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 9.  | C2_S3 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 10. | R1_S1 | Generic                | Ideal       | Res= 13300.0ohm<br>Tolerance= 0.5%             | 1   |
| 11. | R1_S2 | Generic                | Ideal       | Res= 15800.0ohm<br>Tolerance= 0.5%             | 1   |
| 12. | R1_S3 | Generic                | Ideal       | Res= 11100.0ohm<br>Tolerance= 0.5%             | 1   |
| 13. | R2_S1 | Generic                | Ideal       | Res= 26400.0ohm<br>Tolerance= 0.5%             | 1   |
| 14. | R2_S2 | Generic                | Ideal       | Res= 64200.0ohm<br>Tolerance= 0.5%             | 1   |
| 15. | R2_S3 | Generic                | Ideal       | Res= 45300.0ohm<br>Tolerance= 0.5%             | 1   |
| 16. | R3_S1 | Generic                | Ideal       | Res= 1150.0ohm<br>Tolerance= 0.5%              | 1   |

| #   | Name  | Manufacturer | Part Number | Properties                       | Qty |
|-----|-------|--------------|-------------|----------------------------------|-----|
| 17. | R3_S2 | Generic      | Ideal       | Res= 649.0ohm<br>Tolerance= 0.5% | 1   |
| 18. | R3_S3 | Generic      | Ideal       | Res= 459.0ohm<br>Tolerance= 0.5% | 1   |

Sensitivity Analysis

| #  | Name | Series | Tolerance |
|----|------|--------|-----------|
| 1. | Cap  | E48    | 1%        |
| 2. | Res  | E192   | 0.5%      |



Design Inputs

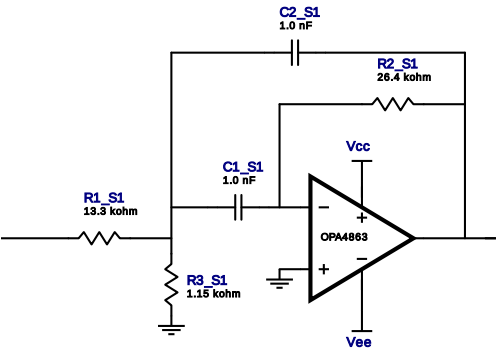
| #   | Name                | Value             | Description                                       |
|-----|---------------------|-------------------|---------------------------------------------------|
| 1.  | FilterType          | bandpass          |                                                   |
| 2.  | FilterResponse      | Butterworth       |                                                   |
| 3.  | FilterOrder         | 6.0               |                                                   |
| 4.  | FilterTopology      | Multiple Feedback |                                                   |
| 5.  | NumberOfStages      | 3.0               |                                                   |
| 6.  | CenterFrequency     | 30.0 k            |                                                   |
| 7.  | StopbandAttenuation | -60.0             |                                                   |
| 8.  | PassbandBandwidth   | 12.0 k            |                                                   |
| 9.  | StopbandBandwidth   | 120.0 k           |                                                   |
| 10. | Gain                | 1.0               |                                                   |
| 11. | DualSupply          | +/-3.30 V         | Power supply(s) to active chips                   |
| 12. | ResistorTolerance   | E192              | Resistor series - 0.5% Passive resistor tolerance |
| 13. | CapacitorTolerance  | E48               | Capacitor series - 1% Passive capacitor tolerance |

Design Assistance

1. **OPA4863** Product Folder : <http://www.ti.com/product/OPA4863> : contains the data sheet and other resources.

Filter Stage :1

Cutoff Frequency      30.108 kHz  
Min GBW Req'd        7.5 MHz  
Stage Gain            992.481 mV/V  
Stage Q                2.497  
Stage Topology        Multiple Feedback

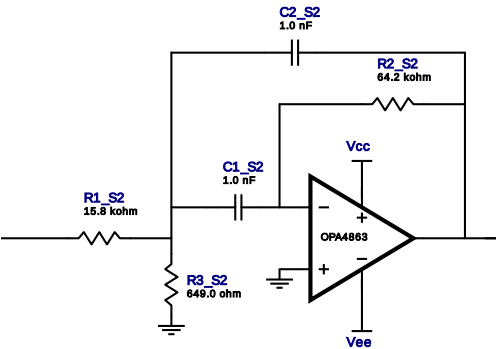


Electrical BOM

| #  | Name  | Manufacturer           | Part Number | Properties                                     | Qty |
|----|-------|------------------------|-------------|------------------------------------------------|-----|
| 1. | A1_S1 | Texas Instruments Inc. | OPA4863     | GbwTyp= 50MHz<br>VccMax= 12.6V<br>VccMin= 2.7V | 1   |
| 2. | C1_S1 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 3. | C2_S1 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 4. | R1_S1 | Generic                | Ideal       | Res= 13300.0ohm<br>Tolerance= 0.5%             | 1   |
| 5. | R2_S1 | Generic                | Ideal       | Res= 26400.0ohm<br>Tolerance= 0.5%             | 1   |
| 6. | R3_S1 | Generic                | Ideal       | Res= 1150.0ohm<br>Tolerance= 0.5%              | 1   |

# Filter Stage :2

Cutoff Frequency 25.158 kHz  
Min GBW Reqd 25.993 MHz  
Stage Gain 2.032 V/V  
Stage Q 5.074  
Stage Topology Multiple Feedback

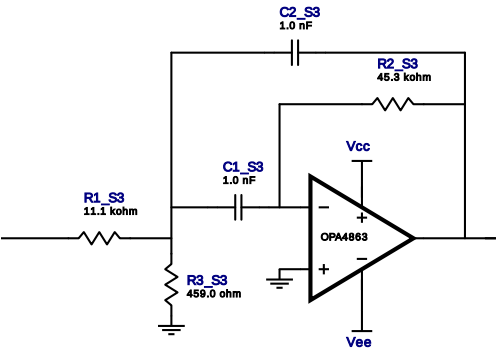


## Electrical BOM

| #  | Name  | Manufacturer           | Part Number | Properties                                     | Qty |
|----|-------|------------------------|-------------|------------------------------------------------|-----|
| 1. | A1_S2 | Texas Instruments Inc. | OPA4863     | GbwTyp= 50MHz<br>VccMax= 12.6V<br>VccMin= 2.7V | 1   |
| 2. | C1_S2 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 3. | C2_S2 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 4. | R1_S2 | Generic                | Ideal       | Res= 15800.0ohm<br>Tolerance= 0.5%             | 1   |
| 5. | R2_S2 | Generic                | Ideal       | Res= 64200.0ohm<br>Tolerance= 0.5%             | 1   |
| 6. | R3_S2 | Generic                | Ideal       | Res= 649.0ohm<br>Tolerance= 0.5%               | 1   |

Filter Stage :3

Cutoff Frequency 35.617 kHz  
Min GBW Req'd 36.752 MHz  
Stage Gain 2.041 V/V  
Stage Q 5.069  
Stage Topology Multiple Feedback



Electrical BOM

| #  | Name  | Manufacturer           | Part Number | Properties                                     | Qty |
|----|-------|------------------------|-------------|------------------------------------------------|-----|
| 1. | A1_S3 | Texas Instruments Inc. | OPA4863     | GbwTyp= 50MHz<br>VccMax= 12.6V<br>VccMin= 2.7V | 1   |
| 2. | C1_S3 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 3. | C2_S3 | Generic                | Ideal       | Cap= 1.0 nF<br>Tolerance= 1.0 %                | 1   |
| 4. | R1_S3 | Generic                | Ideal       | Res= 11100.0ohm<br>Tolerance= 0.5%             | 1   |
| 5. | R2_S3 | Generic                | Ideal       | Res= 45300.0ohm<br>Tolerance= 0.5%             | 1   |
| 6. | R3_S3 | Generic                | Ideal       | Res= 459.0ohm<br>Tolerance= 0.5%               | 1   |

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